A graph-based framework for quantitative analysis of navigational complexity in indoor orienteering courses

July 15, 2026

Abstract

Indoor orienteering presents a navigational challenge distinct from traditional forest orienteering: multi-floor venues, one-way passages, and discontinuous map segments create a dense space of route choices within a small physical area. This article introduces a graph-based framework for modeling indoor venues and courses, defines a notion of route identity based on visible stair crossings that makes exhaustive route enumeration tractable, and describes a first difficulty scoring model calibrated against published course-setting guidelines. The framework is implemented in an open toolchain (web application, command-line tools, and analysis libraries). We position this work as a foundation: the model, the enumeration algorithms, and the collected data enable follow-up research on difficulty scoring models, route-choice behavior, and course design.

1 Introduction

Indoor orienteering is a growing discipline in which competitors navigate multi-floor buildings using specially drawn maps, typically with one map segment per floor printed side by side. Compared to forest orienteering, the navigational difficulty is concentrated in a small area: route choice is dominated by *which staircases to use* and constrained by one-way passages, and the branching factor of decisions per minute is high.

Course setters today rely on qualitative difficulty guidelines, such as the five-level color scale published by Stockholm Indoor Cup [3], and on personal experience. There is no established quantitative method for assessing how difficult a leg or a course actually is, nor for comparing difficulty across venues and events.

This article introduces a framework that makes such analysis possible. The contributions are:

1. a **graph model** of indoor venues that captures corridors, stairs, one-way constraints, and floor membership (Section 3);
2. a definition of **route identity** — two paths are the same route if and only if they cross the same labeled stairs in the same order — which reduces route enumeration from an intractable all-simple-paths problem to a tractable search over label sequences (Section 4);
3. a first **difficulty scoring model**, additive over stair crossings and branching points, mapped onto the established difficulty categories (Section 5);
4. an **open implementation** of the above, together with venue graphs, courses, and anonymized split-time data from real events for future validation (Section 6).

We deliberately keep the scoring model simple and inspectable. The goal of this first article is not a definitive difficulty measure, but a shared vocabulary, data model, and algorithmic toolkit on which such measures can be built and tested.

2 Terminology

Indoor orienteering maps contain structural elements with no direct outdoor counterpart. We use the following terms throughout.

- **One-way passages:** areas where movement is permitted in only one direction, marked on the map.
- **Through passages** (also *square passages/morakvadrat*): crossing passages where movement is permitted only along marked lines; crossing lines do not connect. These may be two-way or one-way.
- **Map-segment transitions:** discontinuous transitions on the paper map, where physically adjacent points appear on different map segments — most commonly floor changes via stairs, but also transitions between segments of the same floor on large venues with multiple buildings.
- **Stair labels:** each staircase carries a unique identifier printed on the map, used both for navigation and for describing routes.

2.1 Describing routes by stair sequences

Stair labels induce a compact route notation: a route is written as the sequence of staircases it uses, with capital letters for upward and lowercase letters for downward traversals. For example, *dcccCA* describes a route that first descends staircase D, descends three floors in staircase C, then ascends C and finally A. The route *within* each floor is normally easy to solve; the essential difficulty of indoor route choice lies in selecting the stair sequence. This observation motivates the definition of route identity in Section 4.

2.2 Difficulty categories

Stockholm Indoor Cup publishes five qualitative difficulty levels [3], summarized in Table 1. These serve as the target scale for quantitative scoring.

Category	Characterization
White	Same floor or very obvious floor changes.
Yellow	Max. one floor change per control.
Orange	Max. three stairs per control; real route choices.
Purple	High complexity, but shorter legs than Black.
Black	Maximum complexity; many stair/one-way choices.

Table 1: Difficulty categories published by Stockholm Indoor Cup (abridged).

3 The venue graph model

We model a venue as a directed multigraph $G = (V, E)$ drawn over the competition map.

Nodes $v \in V$ are decision-relevant positions (junctions, stair entries and exits, passage endpoints), each with a position in map coordinates (millimeters on paper). Control points — start, finish, and numbered controls — are merged into the node set, so pathfinding treats them as ordinary nodes. Nodes may also be inserted at points with no navigational decision, such as corridor corners, purely to improve the accuracy of distance calculations along non-straight paths.

Edges $e \in E$ represent traversable connections and carry three properties that drive all subsequent analysis:

- a **label** $\ell(e)$: non-empty for stairs and ramps (the identifier printed on the map, with separate labels per traversal direction, e.g. upper-case "A" when ascending upstairs, and lower-case "a" when descending downstairs); empty for ordinary corridors;
- a **one-way flag**: if set, the edge is traversable only in its forward direction;
- an **ignore-length flag**: set for map-segment transitions (typically stairs), whose length is not measured.

Floors numbers are not stored on nodes, they can be calculated. One method is to use a seed node with floor number, and then calculate the floor for all other nodes using the labels. We can also define polygons for each floor on the map, and use point-in-polygon lookup to find the floor for each node. These polygons can also be used for 3D visualization of the map.

Distances. With map coordinates in millimeters and a known map scale, the real-world length of an edge can be calculated using the Euclidean distance between its endpoints, scaled to meters.

3.1 Graph notation

Figures in this framework use the following notation.

4 Routes: identity, shortest paths, enumeration

4.1 Route identity

Enumerating all simple paths between two controls is exponential and mostly meaningless:

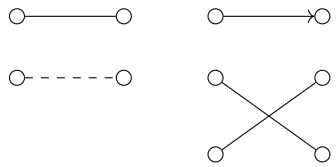


Figure 1: Edge notation. Top left: two-way corridor. Top right: one-way passage. Bottom left: labeled map-segment transition (stair), drawn dashed. Bottom right: through passage — movement only along the marked diagonals, which do not connect where they cross.

two paths that wiggle through different corridors of the same floor are easily identified, and for orienteering purposes we can consider them the *same route*. We therefore define:

> Two paths from control A to control B are the same **route** if and only if they cross the same labeled edges (stairs) in the same order.

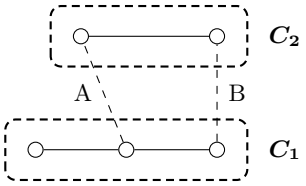
The identity of a route is its *label sequence* — precisely the stair notation of Section 2. The enumeration problem becomes: find all distinct label sequences from A to B, and for each, the shortest concrete path realizing it.

4.2 Cluster condensation

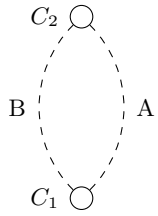
Bidirectional unlabeled edges never contribute to route identity: within a flat region, one can always take the cheapest corridor path. The analysis therefore collapses every connected component of the bidirectional-unlabeled subgraph into a single *cluster* (using a disjoint-set structure). One-way unlabeled edges are excluded from the merge — collapsing them would erase their directionality — and instead become directed edges of the condensed graph.

The resulting **cluster graph** has one node per connected region and one directed edge per traversable direction of each stair or one-way passage (Figure 2). Each physical staircase may

be used at most once per route (in either direction); this single constraint bounds the search space while still allowing a route to revisit the same floor via different stairs — a real phenomenon in multi-floor venues.



(a) venue graph



(b) condensed cluster graph

Figure 2: Cluster condensation. The corridors of each floor collapse into clusters C_1 and C_2; the condensed graph (b) has exactly two edges, stairs A and B. There are two distinct routes $C_1 \to C_2$: label sequences $[A]$ and $[B]$, regardless of which corridors are walked.

4.3 Algorithms

Four algorithm families operate on this model; full derivations and implementation notes are given in the repository's algorithm reference [4].

Shortest path. Dijkstra's algorithm over the venue graph, with edge weights as defined above.

Heuristic alternatives. A k-shortest-paths heuristic in the family of penalty methods: edges on previously found paths are made progressively more expensive (level-change edges $\times 5.0$ per iteration, corridors $\times 1.2$) and Dijkstra is re-run, diversifying stair choice aggressively. Fast, but with no completeness guarantee.

Exhaustive enumeration. A best-first search over the cluster graph enumerates *all* distinct label sequences, shortest first. An exact, admissible heuristic — one Dijkstra run on the reversed full graph from the target — makes the search pop states in non-decreasing order of true route length, so the first completion found for a given label sequence is its shortest realization, and deduplication is a constant-time set lookup. Depth, result-count, and distance-ratio guards keep the (theoretically exponential) search practical; real venues have on the order of 10–20 stair edges and produce route inventories in the hundreds.

Human-route simulation. A fourth algorithm deliberately does not seek the optimal route: it simulates the bounded, satisficing search a human competitor performs under map-reading and working-memory constraints, on the same cluster graph. It short-circuits legs solvable within a single cluster, otherwise runs an alternating bidirectional search that expands the nearest, goal-directed stair on each side in turn and stops at the first meeting point, without backtracking to consider alternatives. The result is a single, non-optimal, deterministic route that can be compared against the enumerated optimum, giving measures — such as the human–optimal length gap — that a purely optimal analysis cannot provide. This algorithm and its cognitive motivation are developed in a companion article [5].

5 Difficulty scoring

5.1 Architecture

Difficulty scoring is isolated behind a pluggable *scoring model* interface: a model receives the graph, and a concrete route, and returns a score. This separation is deliberate — the framework is designed to host competing models so they can be compared against the same route inventories and empirical data.

5.2 The difficulty model

The first model is intentionally simple: a linear combination of two structural counts,

$$D \;=\; k\big(5\,N_f + 3\,N_b\big), \qquad k = \frac{100}{81}$$

where, for the scored route,

- N_f is the number of stair (floor-change) traversals, upward and downward combined (an edge counts as a stair when its endpoints resolve to different floor levels);
- N_b is the number of *branching points*: route nodes at which more than one next node is reachable (excluding the node just come from, and respecting one-way constraints);
- $k = 100/81$ is a scaling factor that normalizes a reference leg (Stockholm Indoor Cup 2023, stage 1, course 1, leg 3→4; raw score 81) to exactly 100.

The term weights, 5 for a stair crossing and 3 for a branching point, were assigned arbitrarily, by judgment rather than by fitting to empirical data; only the scaling factor k is calibrated, and only against a single reference leg. Establishing whether these particular weights are justified is exactly the empirical validation work described in Section 7.

Leg difficulty is the difficulty score of the leg's shortest route; course difficulty defaults to the *maximum* leg difficulty (a course is as hard as its hardest leg), with the mean reported as a secondary statistic. Scores map onto the five-color scale at fixed thresholds (Table 2).

Category	Score range
White	0–20
Yellow	20–40
Orange	40–60
Purple	60–80
Black	80+

Table 2: Difficulty scores mapped to difficulty categories.

5.3 Refinements

Raw branching counts every fork, including trivially dismissible ones. Two refinements are computed alongside the score and available to future models:

- **Dead-end filtering:** edges on non-branching chains that terminate at a leaf are excluded — a passage with only one way in and out creates no real decision.
- **Decision points:** a stricter variant that counts a node only when an off-route branch is *significant*, i.e. runs at least 50 m before dead-ending or rejoining the route. A 10 m alcove is visibly wrong and creates no decision; a 50 m parallel corridor is a genuine route choice.

5.4 Known limitations

This model is a baseline, and its limitations chart the research agenda: branching is counted, not weighted by how hard the alternatives are to reject; the additive form cannot express the combinatorial explosion of several *sequential* decisions; and the score depends on how densely the navigation graph is drawn, so graph-drawing conventions must be standardized before scores are comparable across venues.

6 Implementation and data

The framework is implemented as an open-source toolchain [4]: a client-side web application for drawing venue graphs over competition maps, setting courses, and inspecting route inventories and difficulty scores; and command-line tools for batch analysis, format conver-

sion (JSON, IOF XML 3.0, CSV, GPX), and video visualization of the route-finding algorithms. Venue graphs, courses, and analysis results use documented schemas so that datasets are portable.

The repository also contains anonymized split-time data from real indoor events (per-leg times from public results services), joined with computed per-leg difficulty scores. This dataset is the basis for the validation work described next; no validation results are claimed in this article.

7 Research directions

The framework is a foundation. Directions we consider promising, roughly in order of dependency:

1. **Empirical validation.** Correlate per-leg difficulty scores with observed performance (time loss relative to the leg winner, spread of split times, and mistake frequency) across the collected event data, per difficulty category and runner class.

2. **Model competition.** Fit alternative scoring models (e.g. weighting decision points by branch significance, multiplicative or entropy-based forms) against the same data, using the pluggable model interface, and compare predictive power.

3. **Route-choice entropy.** The exhaustive route inventory enables information-theoretic measures: a leg where several routes have near-equal length carries more decision load than one with a single dominant route. Quantify this and test it as a model term.

4. **Trap detection.** Compare the geometrically shortest route with the lowest-difficulty route; large divergence signals course-setter "baits" worth flagging automatically.

5. **Human route modeling.** Predict which of the enumerated routes runners actually take, and how the choice depends on experience level.

6. **Graph-drawing norms.** Establish conventions for node and edge placement density so difficulty scores are comparable across venues and mappers.

7. **Training effects.** Study whether targeted practice shifts the difficulty–performance relationship.

8 Conclusion

We have introduced a graph-based framework for indoor orienteering course analysis: a venue model capturing stairs, one-way constraints, and floor membership; a definition of route identity via stair-label sequences that makes exhaustive route enumeration tractable; and a first, deliberately simple difficulty scoring model aligned with the established difficulty categories. All components are implemented in an open toolchain. The intended contribution is a foundation: a shared vocabulary and testable baseline against which better models of indoor navigational difficulty can now be built.

References

[1] Indoor-projektet (2023). *Norm för banläggning inom Indoor i Sverige.*

[2] Johan Wettergren (2024). *Råd till banläggare av inomhusorientering.*

[3] Stockholm Indoor Cup. *Svårighetsgrader — Difficulty levels.*

[4] *Indoor Orienteering Route Analysis* (source code, algorithm reference, and datasets). `github:objorke/indoor-o-route-analysis`.

[5] *How humans solve indoor orienteering courses*. Companion article, same repository.